

25) Show that at least ten of any 64 days chosen must fall on the same day of the week.

Solution: Suppose, for the sake of contradiction, at most nine of any 64 days chosen falls on the same day of the week. Since, there are 7 days in a week, and each day can get allotted at most 9 of any 64 days, we have a total of $9 \times 7 = 63$ possible days chosen. This is a contradiction to the fact that we are given 64 days.

$\therefore$ At least ten of any 64 days chosen must fall on the same day of the week. $\square$

26) Show that at least three of any 25 days chosen must fall in the same month of the year.

Solution: Suppose, for the sake of contradiction, at most 2 of any 25 days chosen must fall in the same month of the year. Since, there are 12 months in the year, and each month can get allotted two days, we have a total of $2 \times 12 = 24$ days chosen. This is a contradiction to the fact that we have chosen 25 days.

$\therefore$ At least three of any 25 days chosen must fall in the same month of the year. $\square$

27) ^(V.E. Problem) Use a proof by contradiction to show that there is no rational number r for which $p^2 + r^2 + 1 = 0$.

Solution: $p^2 + r^2 + 1 = 0$, for some $p \in \mathbb{R}$ $q: p$ is an irrational no.

We shall prove $(p \rightarrow q)$ is True by contradiction.

$\therefore$ We shall prove $(p \wedge \neg q) \rightarrow F$ is True.

Let, $(p \wedge \neg q)$ be True. By defn. of AND, p is True and $\neg q$ is True

$\neg q: p$ is not irrational, i.e. p is rational.